

CS4023D Artificial Intelligence

Assignment – 1

Group Name : MindMesh

3-SAT problem

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Problem Statement

3-SAT (3-Satisfiability) is a well-known problem in computational complexity theory and is one of the classic examples of an **NP-Complete** problem. It belongs to the broader family of **SAT (Satisfiability)** problems.

SAT stands for **Boolean Satisfiability**. Given a Boolean formula expressed in **Conjunctive Normal Form (CNF)**—that is, a conjunction (AND) of clauses, where each clause is a OR of literals—the goal is to determine whether there exists an assignment of truth values (true/false) to the literals such that the entire formula evaluates to **true**.

3-SAT is a special case where each clause has 3 literals and is in CNF. The problem statement is as follows:

“Given a Boolean formula in 3-CNF form, determine whether there exists an assignment of truth values to the literals so that the formula evaluates to true.”

State Space Representation

The state space of the problem is represented as a 4-tuple, (N, A, S, GD) which is defined as follows:

- **N (Set of States)**

We assume that our Boolean formula has exactly n literals which are

$x_1, x_2, x_3, \dots, x_n$ and since each clause has 3 variables there will be $n / 3 = k$ clauses.

$$N = \{ (x_1, x_2, \dots, x_n) \mid \forall i \in [1, n], x_i \in \{0, 1, -\} \}$$

Where:

0 $\rightarrow$ Literal assigned **false**

1 $\rightarrow$ Literal assigned **true**

- $\rightarrow$ Literal is **unassigned**

Given a Boolean formula with n literals dimension of N would be 3^N

- **A (Set of Actions)**

There are 2 possible actions from any given state. Choose a literal x_i (un-assigned) and assign it the truth value **true(1)** or **false(0)**.

At level 1 we assign 1 or 0 to x_1 similarly at level 2 we assign 1 or 0 to x_2 like that we proceed.

Assign true

We are at level i , $(x_1, x_2, x_3, \dots, x_i, \dots, x_n) \rightarrow (x_1, x_2, x_3, \dots, 1, \dots, x_n)$

Assign false

We are at level i , $(x_1, x_2, x_3, \dots, x_i, \dots, x_n) \rightarrow (x_1, x_2, x_3, \dots, 0, \dots, x_n)$

- **S (Set of Start State(s))**

Initially all the literals in the Boolean formula will be un-assigned.

$S = \{(x_1, x_2, \dots, x_n) : \forall i \in [1, n], x_i = -\}$

- **GD (Set of Goal State(s))**

We say that a formula is satisfiable if it has is true meaning all the clauses are true.

Let $c_1, c_2, \dots, c_k$ be our **clauses**

$GD = \{(c_1, c_2, \dots, c_k) : \forall i \in [1, k], c_i = 1\}$ and $k = n / 3$.